

Online Neural Space Time Memory for Dynamic Novel View Synthesis

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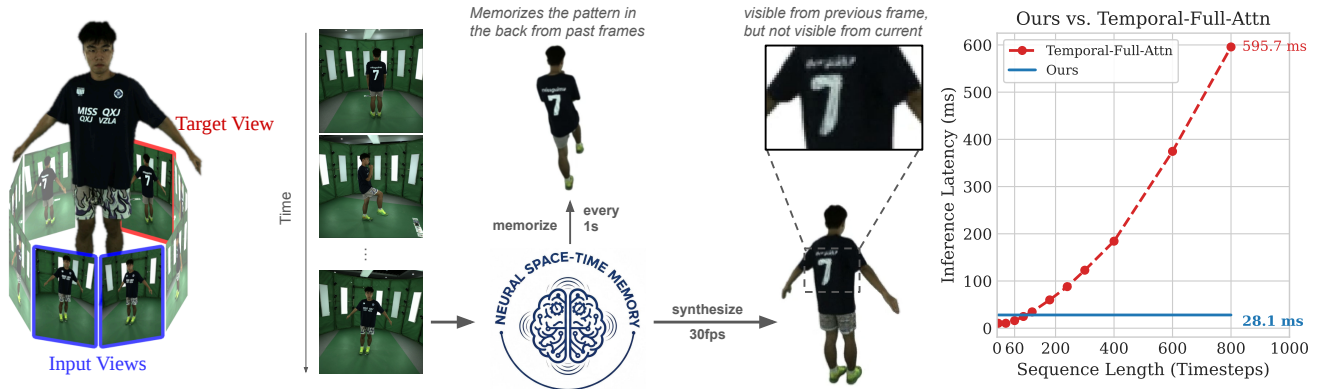


Figure 1. **Neural Space-Time Memory (NSTM)** takes multi-view video streams as input and performs real-time novel view synthesis with minute-long memory. Even when the subject’s back is only observed at $T=0$ and becomes occluded, NSTM faithfully recalls it by fusing current observations with persistent memory.

Abstract

001 Novel view synthesis from dynamic streaming video faces
 002 a fundamental trade-off: maintaining a persistent, long-
 003 horizon memory to reconstruct temporarily occluded re-
 004 gions while operating under strict real-time constraints.
 005 While Test-Time Training (TTT) offers a powerful mem-
 006 ory mechanism, standard models mandate gradient-based
 007 memory updates at every frame to adapt to the chang-
 008 ing motion in dynamic scenes. This computational over-
 009 head from heavy memory updates precludes real-time ap-
 010 plication and can lead to instability over extended con-
 011 texts. Given that memory updates are considerably more
 012 demanding than memory application and that video content
 013 is largely redundant, we propose to decouple the frequen-
 014 cies of these two process. Our approach performs peri-
 015 odic memory updates while applying the memory on a per-
 016 frame basis, using cross-view attention to manage deforma-
 017 tions between the prior memory state and the current frame.
 018 To lock in the historical context, we introduce two critical
 019 mechanisms: an auxiliary Memory Loss that forces per-
 020 sistent internalization of the scene, and a Memory Caching
 021 strategy that regularizes active weights against catastrophic
 022 drift. Our method demonstrates real-time, state-of-the-art
 023 performance on scenes with dynamic human motion as well

as minute-scale online memorization.

1. Introduction

Online novel view synthesis (NVS) on dynamic scenes is a long-standing goal in computer vision, and enables downstream applications such as 3D telepresence and live free-viewpoint broadcast. However, sparse camera setups and occlusions make perceiving an entire scene at a single instant challenging. Spatial memory and reasoning is needed to recall former observations of currently-occluded regions. This ability to reason about object permanence (space-time memory) is especially helpful in the context of novel view synthesis, as it improves the ability of a model to reconstruct a scene at any given moment. However space-time memory methods tend to either degrade over longer time horizons, or scale compute with context size to avoid forgetting.

In this paper, we present an online dynamic novel view system capable of storing persistent, minute-scale historical context to perform grounded synthesis of momentarily occluded regions, all while operating under strict real-time constraints. The key is a memory mechanism based on Test-Time Training (TTT). We selected TTT for its expressive memory capacity and, crucially, its linear scalability – bypassing the prohibitive quadratic computational cost of full self-attention used in typical feed-forward NVS meth-

ods (Fig. 1). However, while TTT has works well in static 3D environments, deploying it on continuous, [streaming](#) dynamic scenes in real-time poses several challenges, as follows.

Speed. The efficiency of recent TTT models stems from large-chunk processing [25, 66, 68]. Naïvely adapting this to dynamic environments requires updating its memory at every frame to adapt to the changing motion. [However, TTT gradient updates are expensive, taking 58ms at 256x256 resolution on an H100 GPU, precluding their per-frame use for real-time applications.](#) In contrast, memory applications without updates can be accomplished at [27ms \(37 FPS\)](#). Leveraging this computational asymmetry and observing that at streaming frame rates, changes in scene content are small, we propose a framework that decouples the frequency of these two processes. Our approach performs infrequent, periodic, heavy memory updates to capture novel states (*Memorization Step*), while applying the memory continuously on a per-frame basis (*Synthesis Step*). During the Synthesis Step, the attention module attends across current input and target views to capture ongoing motion while querying the memory module for past context. Together, they resolve dynamic deformations between historical states and present observations, unlocking amortized real-time performance.

Memory Persistence and Supervision Balance. [Decoupling synthesis and memorization requires careful design of the training objective. This is because occlusion signals in natural video sequences are sparse, and when they do occur, impact a small fraction of pixels. Jointly training the separate memorization and synthesis tasks on this sparse data distribution results in an underconstrained problem, causing the network to over-rely on processing current, incomplete observation, effectively bypassing the memory module and forgetting the past \(Sec. 4.3\).](#) To resolve this, we disentangle the training objectives and propose an auxiliary *memory loss*. During the *Memorization Step*, we query the memory using only the target view camera rays, deliberately preventing any attention over the current input views. By isolating the readout, we force the memory module to reconstruct the scene entirely from its internalized historical context. Then, during the *Synthesis Step*, we combine the current motion from the cross-view attention and the past context from the memory module to synthesize the final novel view. By explicitly supervising the memory’s independent reconstruction capabilities, we force the [memory module](#) to persistently internalize the scene context.

Long-Term Stability. Continuous memory updates in TTT can lead to instability over long horizons. Our periodic update scheme natively mitigates this by reducing the overall frequency of updates. To further guarantee stability, we adopt a *memory caching* strategy [6] that aggregates historical [memory](#) checkpoints, stabilizing the parameter space to

handle minute-long streams. Crucially, modeling deformation in the Synthesis Step enables memory caching to handle changing motion (Sec. 4.3). Finally, we identify that the standard dot-product inner loss used in [state-of-the-art TTT works](#) [25, 66, 68] causes magnitude inflation over many update steps and [show that we can stabilize the framework](#) by replacing it with an L_2 objective [3, 7].

To demonstrate the robust memorization and generalization of Neural Space-Time Memory (NSTM) [for streaming applications](#), we extensively evaluate our framework on a large-scale dynamic human motion dataset, MVHuman-Net++ [58], [processing the sequences strictly in an online manner](#). NSTM successfully achieves stable, minute-long persistent memorization, maintaining high-fidelity synthesis of dynamic scenes while operating in real time.

Our key contributions are summarized as follows:

- **Real-Time, Long-Context Dynamic NVS:** We propose Neural Space-Time Memory (NSTM), the first online dynamic novel view synthesis framework capable of maintaining minute-long persistent memory while operating in real-time.
- **Decoupled Memorization and Deformation:** We overcome the TTT computational bottleneck by structurally decoupling the frequencies between periodic, heavy memory updates and continuous, transformer-driven per-frame synthesis.
- **Disentangled Memory Supervision:** We introduce an alternating *memorization* and *synthesis* training regime and an auxiliary **memory loss** that explicitly forces the network to internalize persistent scene context, overcoming sparse supervision occlusion signals.
- **Memory Caching & Stability:** We enable stable, minute-long view synthesis by aggregating historical states via **memory caching**.

2. Related Works

Novel View Synthesis. The pursuit of high-fidelity novel view synthesis (NVS) has historically centered on designing robust 3D representations to establish multi-view geometric consistencies. Researchers have developed various structures, including meshes [20, 38, 46], multi-plane images [9, 14–16], neural radiance fields [33, 36, 49, 50, 53, 63, 64], point clouds [1, 48, 52, 59], 3D Gaussian splatting [22, 29, 57], etc. While effective, these representations impose implicit inductive biases that can constrain a model’s flexibility when handling complex scenes that deviate from predefined priors.

Recently, the field has shifted toward more general, data-driven architectures to reduce these geometric constraints [39, 42]. One prominent direction leverages diffusion models [21], utilizing strong 2D generative priors to hallucinate unobserved regions within a 3D-aware framework [34, 35, 41, 55, 65]. In parallel, an emerg-

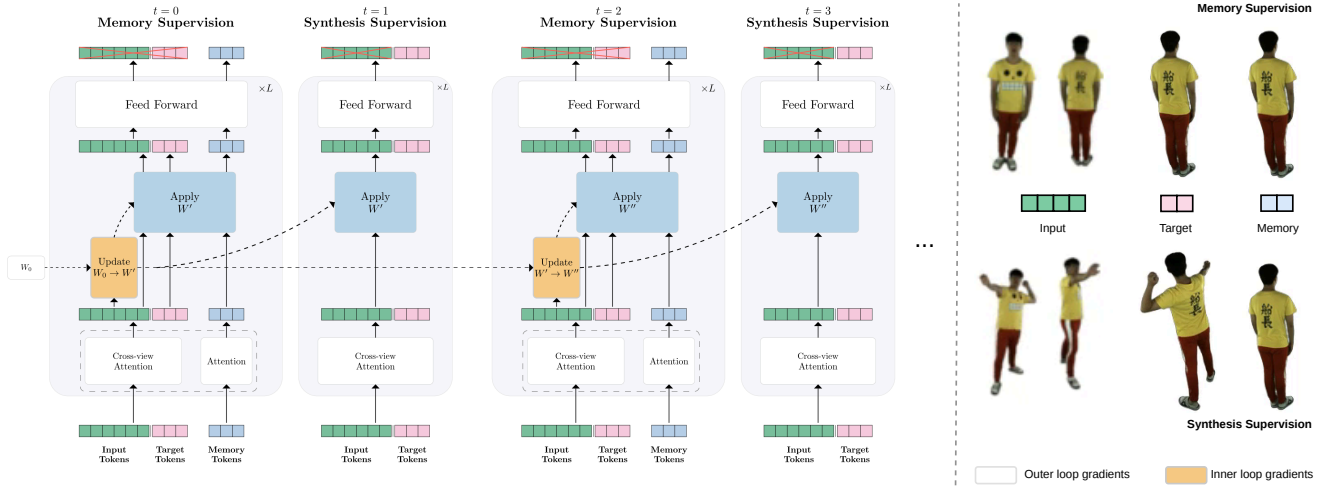


Figure 2. **NSTM Architecture and Training.** **Left:** To prevent the network from bypassing the memory module, we introduce an alternating training schedule. In the *Memory Supervision* step, we introduce isolated memory tokens (target camera rays) that strictly self-attend, preventing any cross-attention with the input views (see Attention Masks, bottom right). After the memory module updates from the input tokens, these isolated tokens query the fast-weights to reconstruct a target view. This forces a readout from the memory alone, ensuring the memory actually internalizes the scene context. In the *Synthesis Supervision* step, input and target tokens freely cross-attend to capture current motion, querying the previously updated memory state without altering it. **Right:** Visualizations of the input, target, and memory from these steps. Crucially, the memory readout in the synthesis step often exhibits a different pose than the current inputs. Our cross-view attention dynamically resolves this spatial deformation, successfully fusing past appearance with current motion.

ing line of research demonstrates that constructing an explicit 3D proxy is not strictly necessary at all. Frameworks such as LVSM [24] and LaCT [68] operate in a purely representation-free manner by formulating NVS entirely as a latent sequence modeling task. Instead of building spatial structures, these methods tokenize source views and utilize expressive transformer architectures to implicitly capture multi-view consistencies. By directly predicting novel views in a feed-forward manner, they successfully bypass traditional 3D structural constraints and highlight the scalability of large vision models.

Streaming Scene Reconstruction. In the context of dynamic environments, streamable novel view synthesis introduces unique complexities, requiring systems to operate causally and efficiently on continuous video streams. Unlike offline dynamic NVS methods that retrospectively optimize over the entire sequence [10, 17, 31, 32, 57, 60, 62], streaming reconstruction demands online updates using only current and historical frames under strict computational budgets. Early streaming approaches [43, 61] rely on per-frame optimization, which is prohibitively slow for real-world applications. To accelerate this process, recent work [54] incorporates optical flow to reuse past information. However, this localized tracking mechanism inevitably accumulates errors over time, leading to long-term degradation. In our work, we address this limitation by encoding a persistent memory directly into the fast weights of the network [68]. This architecture allows our model to faithfully memorize scene geometry and retain historical context, ensuring robust, temporally consistent synthesis

without cumulative drift.

Memory Modules. Transformers [47] are the de-facto backbone of most recent advances in machine learning, ranging from language modeling [8, 56] to computer vision [13] and video generation [24]. In recent years, however, there has been increasing interest in designing alternative sub-quadratic architectures, due to the prohibitive quadratic cost of Transformers on long-context data. A simple but elegant solution is to replace Softmax operator with a separable kernel [28], resulting in a linear attention with a recurrent computation and linear complexity that compresses the context into a fixed-size matrix-valued memory. Building on this concept, several studies explored improving linear attention architectures with incorporating: (1) more powerful separable kernels [2, 3, 27, 28]; (2) more expressive learning rules for better memory management such as Delta [40], Oja’s [23], and Omega [3] rules; (3) different types of forgetting (or retention) gates [7, 19, 37, 44]; and (4) novel hybrid designs of Softmax attention and sub-quadratic architectures [2, 5, 12].

Despite such advances, their fixed-size matrix-valued memory damages the performance of the models in retrieval-based tasks, mainly due to limited capacity of a linear matrix as the hidden state of the recurrent neural network. To this end, a new line of research starts designing methods based on “Test-time Memorization/Training” [7, 45] and more generally Nested Learning [4] that allow the use of: (1) deep memory modules [5, 45] as the state of the recurrent models, where the memory capacity is proven to exponentially grow with the depth of the memory [3]; and

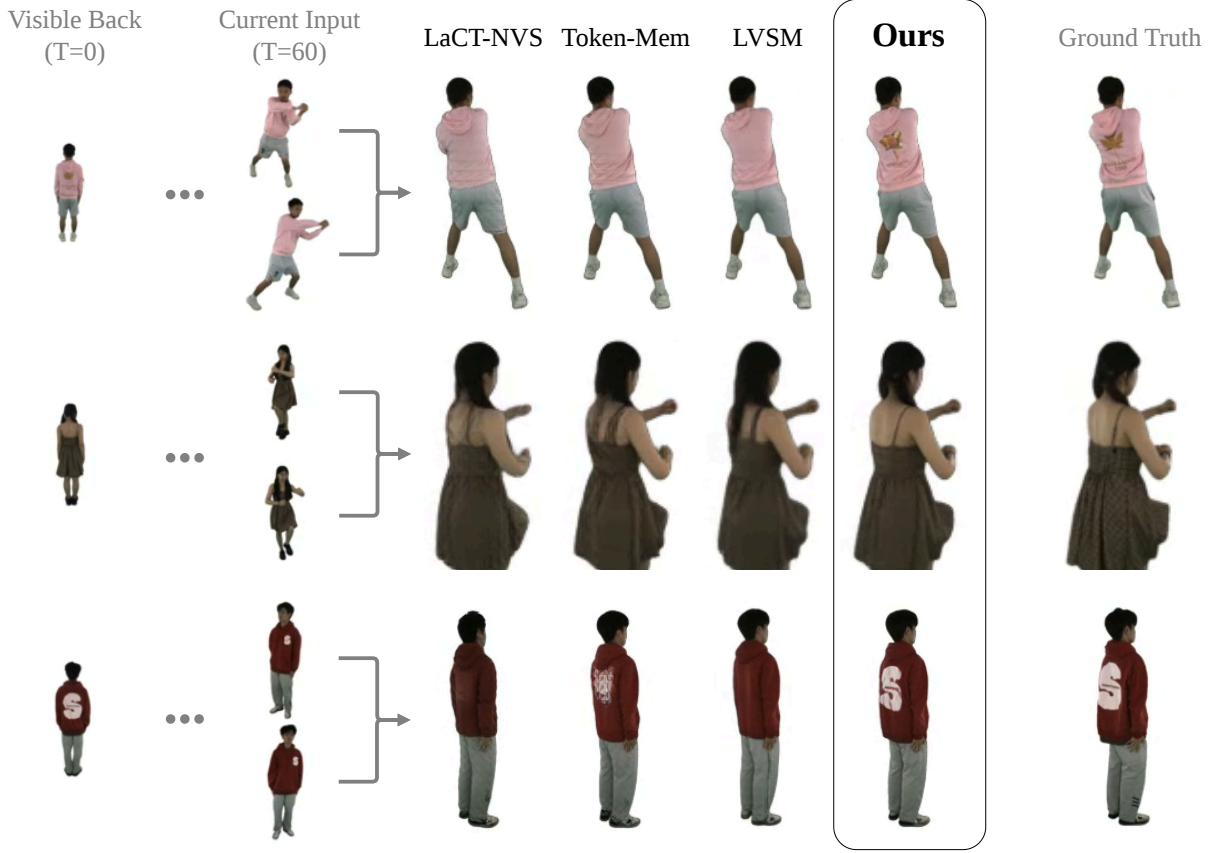


Figure 3. **MVHumanNet++ Qualitative Results.** Comparison with LVSM, Token-Mem (CUT3R), and LaCT-NVS after 1 minute of continuous video. NSTM accurately recalls T-shirt patterns and hair styles. LVSM hallucinates unseen regions, LaCT-NVS suffers OOD errors, and Token-Mem fails to recall back views.

(2) sub-linearly growing memory modules [6, 18] based on structured matrices [18] or memory caching [6], overcoming the fixed-size effective memory of such architectures. Comparing all these memory modules with ours: (1) they are limited to 1-dimensional sequential and so cannot take advantages of rich space-time dependencies in video data; (2) they use a different mechanism of update-apply than ours that causes suboptimal information flow in the time dimension, as validated by our evaluations; (3) contrary to these approaches, our design is based on decoupling memorization and synthesis by using different modules in our architecture; (4) most of them uses a weaker memory management rule based on dot-product inner loss, while we stabilize the memory by taking advantage of L_2 objective.

3. Method

We now describe our method for addressing online novel view synthesis from dynamic video streams. We start with a discussion on Transformer-based novel view synthesis and TTT. Next, Sec. 3.1 describes our decoupled memorization and synthesis architecture, Sec. 3.2 details how NSTM overcomes sparse occlusion supervision signals with our disentangled memory supervision, and Sec. 3.3 describes how we achieve minute-long stable memory.

Transformer-based NVS. At each timestep t , we observe N input views $\mathcal{O}_t = \{(\mathbf{I}_{t,i}, \mathbf{P}_{t,i})\}_{i=1}^N$, where $\mathbf{I}_{t,i} \in \mathbb{R}^{H \times W \times 3}$ is an RGB image and $\mathbf{P}_{t,i} \in SE(3)$ is the corresponding camera pose. Following LVSM [24], we channel-wise concatenate input RGB images with their corresponding per-pixel Plücker ray maps, using linear layers to patchify them into input tokens. Similarly, we patchify the Plücker rays of the target camera to form target tokens. These concatenated input and target tokens are then passed through a stack of Transformer blocks, each comprising an attention module, a TTT memory module (described below), and a feed-forward MLP module. Finally, we extract the processed target tokens and unpatchify them, applying a sigmoid activation to predict the target RGB images (see Sec. D for more details).

Because $\mathcal{O}_t$ provides an incomplete observation of the true dynamic scene (e.g., due to occlusion), NSTM synthesizes the target image $\mathbf{I}_t^{\text{tgt}}$ by conditioning the rendering function Φ on both the current input $\mathcal{O}_t$ and an accumulated implicit memory $\mathcal{W}_{t-1}$:

$$\mathbf{I}_t^{\text{tgt}} = \Phi(\mathbf{P}_t^{\text{tgt}}, \mathcal{O}_t; \mathcal{W}_{t-1}). \quad (1)$$

TTT memory module. Test Time Training/Memorization [7, 45] defines implicit memory as a set of fast weights that

are updated at inference time, forming a compressive neural memory. This contrasts with the *slow weights* (i.e., model parameters) that are frozen during inference. In our architecture, we use $\mathcal{W}_{t-1} = \{W_1, W_2, W_3\}$ as the fast weights with the design of a bias-free SwiGLU-MLP:

$$f_{\mathcal{W}}(x) = (\text{SiLU}(W_1 x) \odot (W_3 x)) W_2, \quad (2)$$

Input tokens are projected into keys k , queries q , and values v , then to read from the weights we **apply** them via $(o = f_{\mathcal{W}}(q))$. The **update** operation memorizes context via a single gradient step on an inner loss binding keys to values,

$$\mathcal{W}' = \mathcal{W} - \eta \nabla_{\mathcal{W}} \mathcal{L}_{\text{inner}}(f_{\mathcal{W}}(k), v). \quad (3)$$

The inner loss is a design choice that distinguishes different existing architectures and determines the memory management in prioritizing the input tokens. In this paper, we use an L_2 inner loss:

$$\mathcal{L}_{\text{inner}} = \|v - f_{\mathcal{W}}(k)\|_2^2. \quad (4)$$

In Tab. 2, Fig. 5 we show that contrary to the choice of unbounded dot-product loss as in some previous models (such as LaCT [68] or DLA [3]) that cause severe magnitude inflation of $f_{\mathcal{W}}(k)$ after repeated gradient descent over long horizons, our choice of L_2 objective results in long-horizon stability.

We further stabilize the update by Newton–Schulz orthonormalization of the gradient (Muon [26]) and L_2 weight normalization. Our network is trained end-to-end via an outer-loop task loss over the slow weights, initial fast-weight state, and projections. We denote the fast-weight state at timestep t as $\mathcal{W}_t$ (Sec. 3.3).

3.1. Decoupled Memorization and Synthesis

Naïvely applying our memory architecture summarized in Sec. 3 to live video streams introduces a latency bottleneck. Standard TTT-based model rigidly couples the memory update and apply steps. To achieve high throughput, prior static frameworks [68] process sequence chunks in parallel. However, this relies on *chunk-level buffering*, which introduces unacceptable delays and violates the strict online constraints of live streaming. Conversely, processing the stream sequentially, by forcing a heavy gradient-based memorization at every incoming frame, incurs prohibitive computational overhead of 58ms (see Tab. 5), severely precluding real-time performance. To overcome this cost, NSTM decouples the frequency of memory updates from novel view synthesis, and decomposes the online inference process into two frequency-asymmetric steps (Fig. 2):

Periodic Memorization Step: Rather than updating at every frame, we strictly limit heavy gradient updates to a periodic schedule to compress novel states into the fast-weights.

Per-frame Synthesis Step: At every timestep t , we execute

a lightweight query against the persistent memory $\mathcal{W}_{t-1}$ Eq. (1). However, because the scene is dynamic and memory updates are periodic, the memory state and the current observation are inherently misaligned (Fig. 2). To handle this temporal deformation, we adopt a *cross-view attention* mechanism. Jointly attending across the source posed images and the target camera rays explicitly captures ongoing motion. The output then queries the memory module, dynamically fusing the current motion with historical context to synthesize the novel view at amortized real-time speeds (Fig. 1). Note that, this process is opposed to the static per-image attention used in recent TTT-based architectures such as ZipMap [25] and LaCT [68].

3.2. Disentangled Memory Supervision

A naïve joint training of the cross-view attention module and the memory module is fundamentally underconstrained. Because disocclusion signals in streaming video are naturally sparse, the model tends to over-rely on the attention and effectively bypasses historical memory. To force the persistent internalization of scene context, we introduce an alternating training regime across a sequence of T timesteps.

Memory Supervision. As illustrated in Fig. 2, on even timesteps ($t = 0, 2, \dots$), we read out from the memory and supervise this readout to reconstruct a random target view. To enforce this loss on the memory alone, it is critical to ensure we read out purely from the memory, completely isolated from the current inputs. Therefore, we introduce additional *memory readout tokens*, representing the camera rays of the target view. Crucially, we restrict the attention mask of the aforementioned cross-view attention module such that these memory tokens only self-attend amongst themselves. These isolated tokens then query the updated memory and pass through the feed-forward layers. Ultimately, they form an image, denoted as $\mathbf{I}_t^{\text{mem}}$, relying entirely on the internalized historical state. We supervise this blind retrieval via an auxiliary *memory loss*:

$$\mathcal{L}_{\text{mem}}^{(t)} = \mathcal{L}(\mathbf{I}_t^{\text{mem}}, \mathbf{I}_t^{\text{gt}}), \quad (5)$$

$$\text{where } \mathcal{L}(\mathbf{I}, \mathbf{I}^{\text{gt}}) = \text{MSE}(\mathbf{I}, \mathbf{I}^{\text{gt}}) + \lambda_{\text{LPIPS}} \text{LPIPS}(\mathbf{I}, \mathbf{I}^{\text{gt}}).$$

$\lambda_{\text{LPIPS}} = 0.5$ is the weight to balance the LPIPS loss [67].

Synthesis Supervision. On odd timesteps ($t = 1, 3, \dots$), the network operates in standard novel view synthesis mode and we supervise the synthesized target view. Here, the standard target ray tokens freely cross-attend with the tokenized inputs $\mathcal{O}_t$ while querying the persistent memory $\mathcal{W}_{t-1}$ inherited from the immediately preceding memorization step. Consequently, despite dynamic deformations between the current observation and the previous memory state, if a region is occluded, the network learns to dynamically fuse and recover it from this preceding memory. We

supervise the output target image from this joint synthesis, denoted as $\mathbf{I}_t^{\text{synth}}$, using the identical loss formulation:

$$\mathcal{L}_{\text{synth}}^{(t)} = \mathcal{L}(\mathbf{I}_t^{\text{synth}}, \mathbf{I}_t^{\text{gt}}). \quad (6)$$

This alternating process continues until all T observations in the sequence are processed. The final training objective minimizes the joint loss across all timesteps, balancing the primary novel view synthesis task against the auxiliary memory regularization:

$$\mathcal{L}_{\text{total}} = \sum_{\substack{t=1 \\ t \text{ is odd}}}^T \mathcal{L}_{\text{synth}}^{(t)} + \sum_{\substack{t=0 \\ t \text{ is even}}}^T \mathcal{L}_{\text{mem}}^{(t)}, \quad (7)$$

where λ_{mem} controls the strength of the memory supervision.

Training-Inference Asymmetry. While we train with a 1:1 alternating ratio, our framework can execute significantly more synthesis steps than memory updates during inference. This discrepancy does not degrade performance because the synthesis steps between two memorization steps read out from the same frozen memory state and are completely independent of one another. See our supplemental videos where we synthesize 29 frames for every 1 step of memorization (29:1).

3.3. Memory Caching

TTT suffers from RNN drift over long-horizon updates. We observe this as memory decay, stemming from optimization instability, in the LaCT-NVS [68] column of Fig. 4. To address this problem, we adapt a memory caching strategy recently proposed for large language models [6, 7].

Concretely, the sequential application of our periodic memory update generates a sequence of evolving memory states $\mathcal{W}_0, \mathcal{W}_1, \mathcal{W}_2, \dots$, where the ones between updates are the same. Note, at any timestep t , the active memory is $\mathcal{W}_{t-1}$ since the memory is not reset after a snapshot is taken. We average over the active memory and every K past memory states, which can be implemented recursively.

$$\bar{\mathcal{W}}_0 = \mathcal{W}_0, \quad \bar{\mathcal{W}}_n = \frac{n-1}{n} \bar{\mathcal{W}}_{n-1} + \frac{1}{n} \mathcal{W}_{nK}, \quad \forall n > 1,$$

$$\hat{\mathcal{W}}_{t-1} = \frac{1}{m+1} \left(\mathcal{W}_{t-1} + \sum_{i=1}^m \mathcal{W}_{iK} \right) = \frac{\mathcal{W}_{t-1} + m \bar{\mathcal{W}}_m}{m+1}$$

where $m = \lfloor (t-1)/K \rfloor$. $\bar{\mathcal{W}}_m$ is the average of every K past states. In practice, we only keep one extra set of weights $\bar{\mathcal{W}}_m$ to track the whole average. This aggregated state $\hat{\mathcal{W}}_{t-1}$ replaces standard memory conditioning (Eq. 1), stabilizing the parameter space and mitigating the out-of-distribution drift caused by continuous gradient updates. Importantly, this memory caching strategy fails on standard, coupled TTT architectures. Its effectiveness fundamentally relies on our cross-view attention to explicitly handle spatial deformations, as we empirically validate in Sec. 4.3.

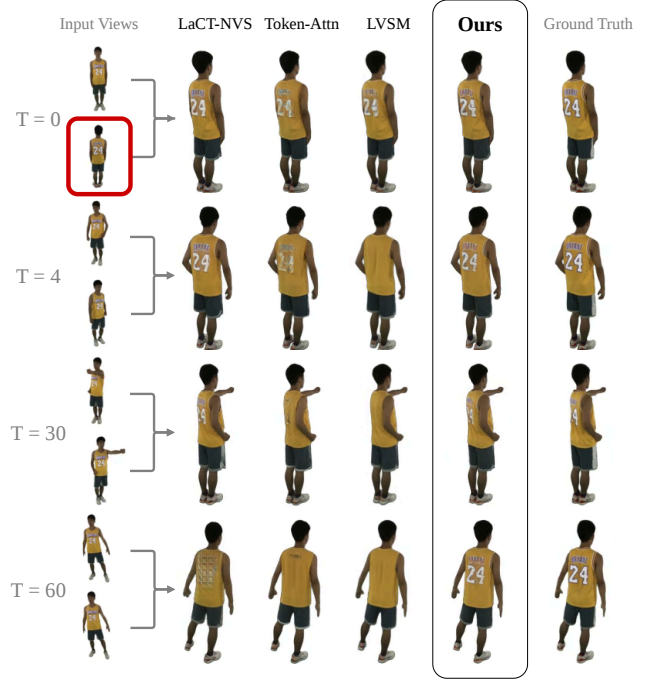


Figure 4. **Memory Stress Test Over Time.** We show the back view at $T=0$. For subsequent timestamps, we show only frontal views and task the model with synthesizing the occluded back. NSTM maintains high-fidelity recall while baseline models suffer from drift or collapse to stateless prior over time.

4. Experiments

We validate NSTM’s ability to maintain real-time, high-fidelity, minute-long memory under severe occlusions. We detail our experimental setup in Sec. 4.1, evaluate memory persistence and synthesis fidelity in Sec. 4.2, deconstruct our architecture via an ablation study in Sec. 4.3, and analyze our amortized efficiency in Sec. 4.4. We urge readers to view our supplementary videos for convincing comparisons.

4.1. Experimental Setup

Dataset. We evaluate on MVHumanNet++ [30], featuring 60,000 multi-view motion sequences across 4,500 identities and 9,000 outfits, using a 90%/10% train/test split. Its wide view coverage, unconstrained motion and minute-long sequence length make it an ideal testbed for dynamic NVS and long-term memory tasks.

Evaluation Protocol. Our dataset comprises 60-frame sequences (subsampling at 1.2 FPS from 50s of raw video), enabling minute-scale evaluation. However, measuring long-term memorization in the wild is challenging as natural disocclusions are spatially and temporally sparse – only 17 of 390 test scenes exhibit full rotations. To comprehensively evaluate memory, we employ two protocols. (1) *Memory Stress Test*: A forced-occlusion setup to evaluate memorization across all scenes. The model observes full front

Method	4 Timesteps						30 Timesteps						60 Timesteps					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	29.13	0.9666	0.0258	0.1040	19.55	0.5790	28.98	0.9656	0.0263	0.1055	19.40	0.5707	28.99	0.9662	0.0257	0.1034	19.34	0.5670
LaCT-NVS	30.69	0.9741	0.0196	0.0822	21.72	0.7110	30.38	0.9708	0.0216	0.0885	21.22	0.6507	25.85	0.9500	0.0360	0.1236	16.25	0.5158
Token-Mem	29.36	0.9674	0.0249	0.1009	20.40	0.6409	28.36	0.9621	0.0290	0.1136	19.14	0.5624	28.11	0.9617	0.0295	0.1141	18.84	0.5505
Ours	30.43	0.9731	0.0203	0.0851	21.46	0.6971	30.03	0.9700	0.0225	0.0915	20.86	0.6434	29.99	0.9701	0.0225	0.0919	20.71	0.6286

Table 1. **Memory Stress Test Comparison.** best, second best, and third best are colored. We test memory retention quality across 360 scenes and three durations as described in Sec. 4.1, taking measurements at the last timestep. While our method maintains a 20+ mPSNR across all durations tested, LaCT-NVS drops by 5.47 dB and Token-Mem by 1.56 dB as the numbers of past observations increases from ($t = 4$) to ($t = 60$). As a stateless model, LVSM fails to reconstruct any view not directly seen in the inputs as shown by close to flat performance across all durations. Please see comprehensive video comparison results in our supplementary website.

and back views at $T = 0$. For subsequent frames, we restrict inputs to frontal stereo views and task the model with synthesizing the occluded back. Evaluations at 4, 30, and 60 frames quantify memory decay from short-term horizons to the minute-scale limit. (2) *Memory from Natural Rotations*: For the 17 full-rotation scenes, we provide fixed frontal stereo inputs and evaluate synthesis as the back naturally reveals and occludes. To avoid the backgrounds dominating the metrics, we report masked mPSNR and mSSIM isolated over the dynamic foreground. Results for the Memory Stress Test are presented in Sec. 4.2 and results for the Memory from Natural Rotations and a standard, unconstrained NVS setup are presented in Sec. B.

Baselines. We compare against the stateless LVSM [24] and two stateful baselines: fast-weight-based LaCT-NVS [68] (adapted to causal updates for dynamic scenes) and token-memory-based Token-Mem (an NVS adaptation of CUT3R [51]). See Sec. A.3 for further adaption details on baseline models. Following their original designs, both stateful methods update memory per frame. Based on our memorization-synthesis decoupling, NSTM updates memory every two frames. Note our subsampled video test cases actually artificially benefit the stateful baselines by mitigating their rapid drift at higher frame rates. In contrast, NSTM’s results accurately reflect true performance on normal high-FPS videos as its dense synthesis steps operate independently between these periodic memorization updates.

Implementation Details. We implement NSTM and all baselines in JAX. All models are trained for 500k iterations using identical optimizer hyperparameters with a total batch size of 128, utilizing 64 NVIDIA H100 GPUs. Real-time inference latency is evaluated on a single H100 GPU. During training, we employ a two stage training curriculum consisting of a memory bootstrapping phase on short $T = 4$ video clips followed by a $T = 24$ free viewpoint and long memory persistence fine tuning phase. Full details are presented in Sec. A.1. Finally, for all models, the perceptual loss weighting hyperparameter is set to $\lambda_{\text{lpiips}} = 0.5$.

4.2. Memory Stress Test

We present our quantitative comparisons in Tab. 1 and qualitative visual evidence in Fig. 3. In the short term ($T = 4$), the advantage of explicit memory is clear: NSTM, LaCT-NVS, and Token-Mem all outperform the stateless LVSM,

Method	Ablation Study (over 60 timesteps)				
	PSNR↑	SSIM↑	LPIPS↓	mPSNR↑	mSSIM↑
LaCT-NVS Baseline	22.32	0.9275	0.0616	13.49	0.4942
LaCT-NVS w/ L_2	27.77	0.9594	0.0275	19.08	0.6189
LaCT-NVS w/ L_2 w/ Mem Caching	20.25	0.9135	0.0688	12.03	0.4707
Ours w/ $\mathcal{L}_{\text{mem}}$	28.23	0.9639	0.0251	19.36	0.6198
Ours w/ Mem Caching	29.24	0.9673	0.0232	19.78	0.6321
Ours w/ Mem Caching	30.09	0.9705	0.0217	20.79	0.6537

Table 2. **Ablation study of our model.** best, second best, and third best are colored. Metrics are averaged over 60 timesteps in 390 MVHumanNet++ [58] eval scenes.

which struggles without historical context. However, the fragility of prior stateful methods becomes visible over the minute-long context ($T = 60$). Token-Mem quickly forgets historical context, its performance comparable to the stateless LVSM that never observed the back view. LaCT-NVS suffers a severe 5.47 dB drop due to the instability of its continuous update mechanism. This quantitative collapse is clearly visible in Fig. 3 and Fig. 4: where LaCT-NVS suffers from out-of-distribution (OOD) drift, while LVSM produces blurry outputs in unseen regions (e.g. the back) due to its stateless nature.

On the other hand, NSTM maintains highly robust long-term memory which is especially pronounced in our masked foreground metrics (mPSNR/mSSIM), proving the model accurately locks in the dynamic subject’s identity rather than relying on the background. Qualitatively (Fig. 3), NSTM faithfully reconstructs specific, temporarily occluded details—such as complex logo and hair styles—long after they disappear from the input stream. This sustained high-fidelity recall directly validates the stabilizing effect of our L_2 objective and memory caching strategy.

4.3. Ablation Study

We validate our design choices in Tab. 2 and Fig. 5, evaluating the impact of each component over 60 timesteps on 390 unseen dynamic scenes from MVHumanNet [58]. **Stability via L_2 Objective.** We first establish a naïve causal baseline (LaCT-NVS) [68] that updates memory at every frame without decoupling memorization from synthesis. As observed, the unbounded dot-product inner loss causes magnitude inflation over continuous updates, resulting in poor performance (22.32 PSNR). Replacing this with a stable L_2 objective (LaCT-NVS w/ L_2) significantly improves stability and synthesis quality (+5.45 PSNR).

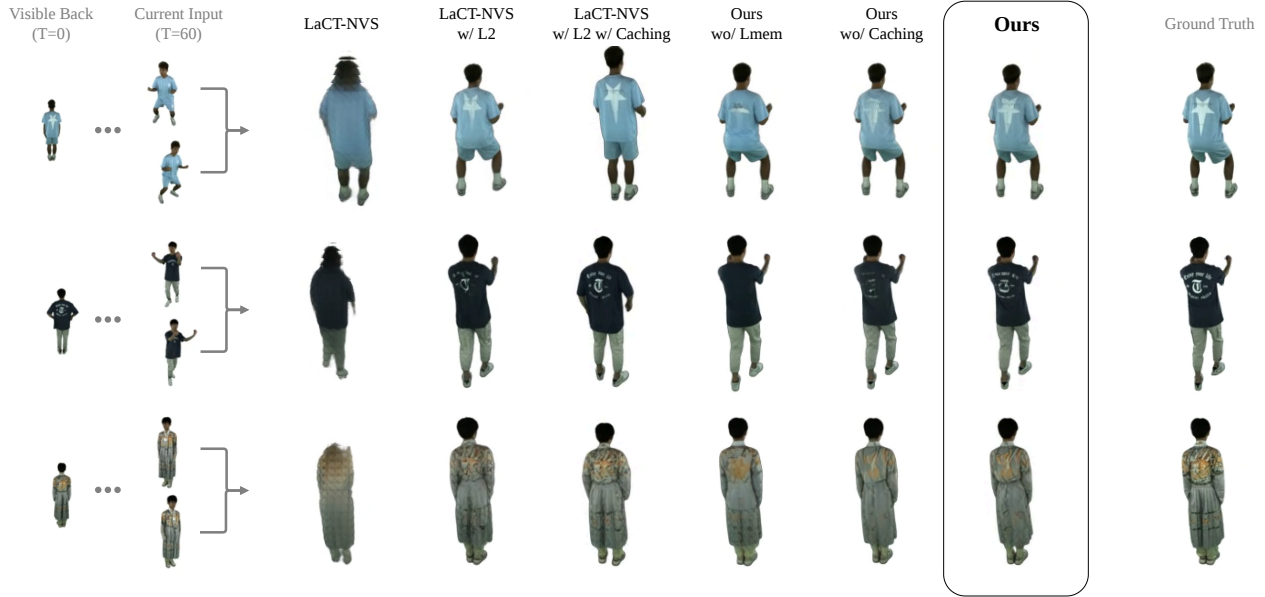


Figure 5. **Qualitative ablation study.** Results at $T = 60$ demonstrate the visual necessity of our decoupled memory framework and stabilization strategies. Only our full framework (Ours) accurately recalls the historical back view with minimal instability.

Memory Caching Requires Decoupled Training. While memory caching is designed for long-term stability, it fundamentally fails on coupled architectures. Row 3 evaluates the LaCT-NVS with L_2 baseline by performing its standard update-and-apply step at every frame, but uniquely conditioning it on the aggregated memory cache (Sec. 3.3). This causes a severe performance drop (20.25 PSNR). The synthesized output becomes visually “stuck” in a historical pose (see LaCT-NVS w/ L_2 w/ caching column in Fig. 5). This occurs because temporal motion is inherently baked into the fast-weights; aggregating these weights creates a superposition of past poses. In our decoupled architecture (Sec. 3.1), the cross-view attention explicitly learns to resolve and deform from this superimposed historical memory to match the current observation. The tightly coupled baseline lacks this explicit deformation mechanism, causing it to simply output the averaged past poses. This proves that structurally decoupling memorization from synthesis is strictly necessary to successfully aggregate historical memory.

Disentangled Supervision Enhances Memory Persistence. When we fully decouple the architecture but remove the auxiliary memory loss $\mathcal{L}_{\text{mem}}$ (Ours wo/ $\mathcal{L}_{\text{mem}}$), performance drops to 28.23 PSNR. Without explicit pressure to reconstruct the scene independently, the network takes a shortcut, over-relying on the cross-view attention’s current visual context and partially collapsing to a stateless prior. Introducing $\mathcal{L}_{\text{mem}}$ forces robust geometric internalization (+1.01 PSNR).

Long-term Stability via Memory Caching. Finally, applying our memory caching strategy to the fully decoupled and properly supervised model (Ours w/ Mem Caching) yields our highest performance (30.09 PSNR), successfully

mitigating long-term weight drift and preserving minute-long context.

4.4. Real-Time Inference Efficiency

To validate our claim of real-time, long-horizon generation, we compare the amortized inference latency of NSTM against Temporal-LVSM, a temporal version of LVSM[24]. We handle long contexts in this model by computing dense attention across all past observations that we want to memorize. As shown in Fig. 1 this imposes a quadratic computational cost, causing latency to spike and precluding real-time rendering on continuous streams. In contrast, NSTM structurally bypasses this bottleneck, achieving constant inference latency regardless of video length. This decoupling successfully severs the tie between sequence length and computational cost, unlocking sustained, real-time dynamic view synthesis.

5. Conclusion and Limitations

In this work, we presented Neural Space-Time Memory (NSTM), a novel framework that bridges the gap between long-context memory and computational efficiency for dynamic novel view synthesis. By disentangling periodic memory updates from per-frame transformer-based deformation modeling, and introducing a targeted memory auxiliary loss, NSTM successfully internalizes persistent scene context while maintaining real-time speed. Furthermore, our memory caching strategy and stable L_2 update rule enhance long-term stability, unlocking minute-long memory horizons and achieving state-of-the-art results compared to existing baselines.

While NSTM provides a principled solution for stream-

ing dynamic novel view synthesis, we identify several avenues for future research. *Update Frequency and Missed Events*: Our method updates the memory periodically. Despite the redundancy of the input videos, the updates may miss events that happen in between. To avoid this issue, one could perform large chunk updates, which updates the memory on multiple frames at a time, each with their own time embedding. This change would also enable the model to perform updates with irregular frame spacing (e.g., unsynced cameras on a multi-view rig). In general, this paper leaves open the question of which frames should be memorized. Future work could explore the tradeoff between more frequent updates and larger-chunk “batch” updates. *Finite Memory Capacity*: Models like ours will eventually hit capacity limits and forget past observations. To improve persistence and stability, future iterations could manage memory caching weights in a learned way. *Memory Primacy Bias*: The method exhibits a bias towards better memorization during early steps. Future methods could explore larger-scale training for longer sequences and better data sampling to inject informative frames in the middle of the sequence, while having good sampling to encourage long-term memorization. *Complex Motion and Trajectories*: Resolving long-range camera motion and significant subject motion within the compressed memory remains challenging and is left for future work.

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A. Additional Implementation Details

In this section, we provide comprehensive details regarding our training curriculum, optimization hyperparameters, baseline configurations, and the evaluation protocol for our generalized Novel View Synthesis experiments.

A.1. Two-Stage Training Curriculum

All stateful models are trained with a two-stage training curriculum. Pretraining bootstraps our model’s memory and is carried out on short video clips consisting of $T = 4$ timesteps. We then finetune for Free-Viewpoint synthesis and Long Memory Persistence with long $T = 24$ video clips. At each timestep, the network processes 2 sparse input views to synthesize 1 target view. Details on each stage follows:

Stage 1: Memory Bootstrapping (Pretraining). The objective of this initial phase is to explicitly encourage the network to build and rely upon its memory module. For the first frame of the sequence ($t = 0$), the input consists of one frontal view (cam00) and one back-facing view (cam08), providing broad geometric context. For subsequent frames ($t > 0$), the inputs transition to two random views. Crucially, the target supervision is a target view that is explicitly distinct from the $t > 0$ inputs. Note these target viewpoints remain *static* across the sequence. This rigid setup boosts memorizing the initial context to faithfully inpaint the targets at later timesteps.

Stage 2: Free-Viewpoint and Long Memory Adaptation (Finetuning). Once the memory module is bootstrapped, we adapt the network to encourage longer memory persistence and enable the memorization of arbitrary, time-varying viewpoints. We extend the pretraining sequence from 4 frames to 24 frames and supervise the network at each timestep by a random, time-varying target view strictly withheld from the input stream. To construct the inputs, two random base cameras are selected and held constant for all $t > 0$. At the initial timestep ($t = 0$), we enforce a forced-occlusion scenario by exclusively sourcing inputs from two distinct, alternative cameras. Finally, to ensure the model properly learns object-centric rotation rather than just camera translation, 50% of the training sequences undergo coordinate alignment. Specifically, we modify the extrinsics of the first initial $t = 0$ camera to match the first base view. Throughout this extended sequence, we maintain the alternating memorization and synthesis steps utilized in the pre-training phase.

A.2. Training Hyperparameters

We implement our framework using JAX. Models are trained with a total batch size of 128 for 500,000 iterations during the pretraining stage, followed by an additional 25,000 iterations during the finetuning stage. We

employ a cosine learning rate decay schedule with a linear warmup. During pretraining, the learning rate is initialized at 1×10^{-6} , warms up over 4,000 steps to a peak of 4×10^{-4} , remains constant for 100,000 steps, and then decays down to a final value of 1×10^{-6} . During the finetuning stage, we retain the same schedule parameters but reduce the peak learning rate to 4×10^{-5} warming up over 4,000 steps and then remaining constant. Finally, for the image reconstruction objective across all models and stages, the perceptual loss weight is fixed at $\lambda_{\text{lips}} = 0.5$.

A.3. Baseline Adaptations and Coordinate Alignments

To ensure a fair comparison, all baselines evaluated in Tab. 1 are trained on the exact same two-training curriculum. However, due to architectural differences, camera coordinate systems must be aligned differently:

LVSM [24]: We utilize the exact architecture described in the original LVSM paper. But for fair comparison, we use the same learning rate schedule as above and the same L2 and LPIPS loss. Because LVSM is a stateless, per-frame model, we construct its training batches by extracting only the final frame ($t = 3$) from our 4-timestep video clips, ensuring it trains on identical data distributions. For coordinate alignment, we relativize the camera coordinate system to each individual target camera over time (i.e., the specific target camera for that frame is set as the origin with an identity rotation).

LaCT-NVS [68] and NSTM (Ours): Stateful models must maintain coherent camera coordinate system across time and adapt to time-varying target cameras. Therefore, for LaCT and NSTM, we relativize all cameras in the sequence to the first target camera ($t = 0$), ensuring a stable, global coordinate space for the fast-weights to operate within.

Token-Mem [51]: As explained in Sec. 4.1, *Token-Mem* is a custom novel view synthesis model whose memory mechanism is based on CUT3R [51]. As a stateful method, it also requires a temporally consistent coordinate frame, and we use the same coordinate system as described above for LaCT and NSTM (ours). Architecture-wise, we follow the original CUT3R [51] implementation to use a state size of 768 tokens each with 768 feature channel. Differing from [51], we optimize L2 and LPIPS losses consistent with the other methods compared.

B. Additional Experiments

B.1. Memory from Natural Rotations Test

To evaluate NSTM on the general setting of natural video sequences where occlusions can happen naturally at any point, we evaluate 17 scenes in MVHumanNet++ that feature natural rotations which show the back of the participants. This is in contrast to our “Memory Stress Test”

Method	Average over T=60						Last frame					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	28.81	0.9654	0.0286	0.1216	19.21	0.5069	28.81	0.9656	0.0296	0.1298	19.69	0.5272
LaCT-NVS	29.13	0.9667	0.0253	0.1045	19.98	0.5850	27.01	0.9551	0.0340	0.1414	17.93	0.4982
Token-Mem	27.24	0.9565	0.0353	0.1303	18.05	0.4926	28.11	0.9615	0.0329	0.1363	19.14	0.5194
Ours	29.53	0.9692	0.0247	0.1056	20.39	0.5928	29.68	0.9691	0.0250	0.1107	20.88	0.5997

Table 3. **Memory from Natural Rotations Test Comparison.** best, second best, and third best are colored. NSTM outperforms Token-Mem and LaCT-NVS by 1.74 dB and 2.95 dB mPSNR on last frame measurement, while LVSM as a stateless model has close to identical average and last frame performance, demonstrating that it is not able to remember past details that would result in increased accuracy.

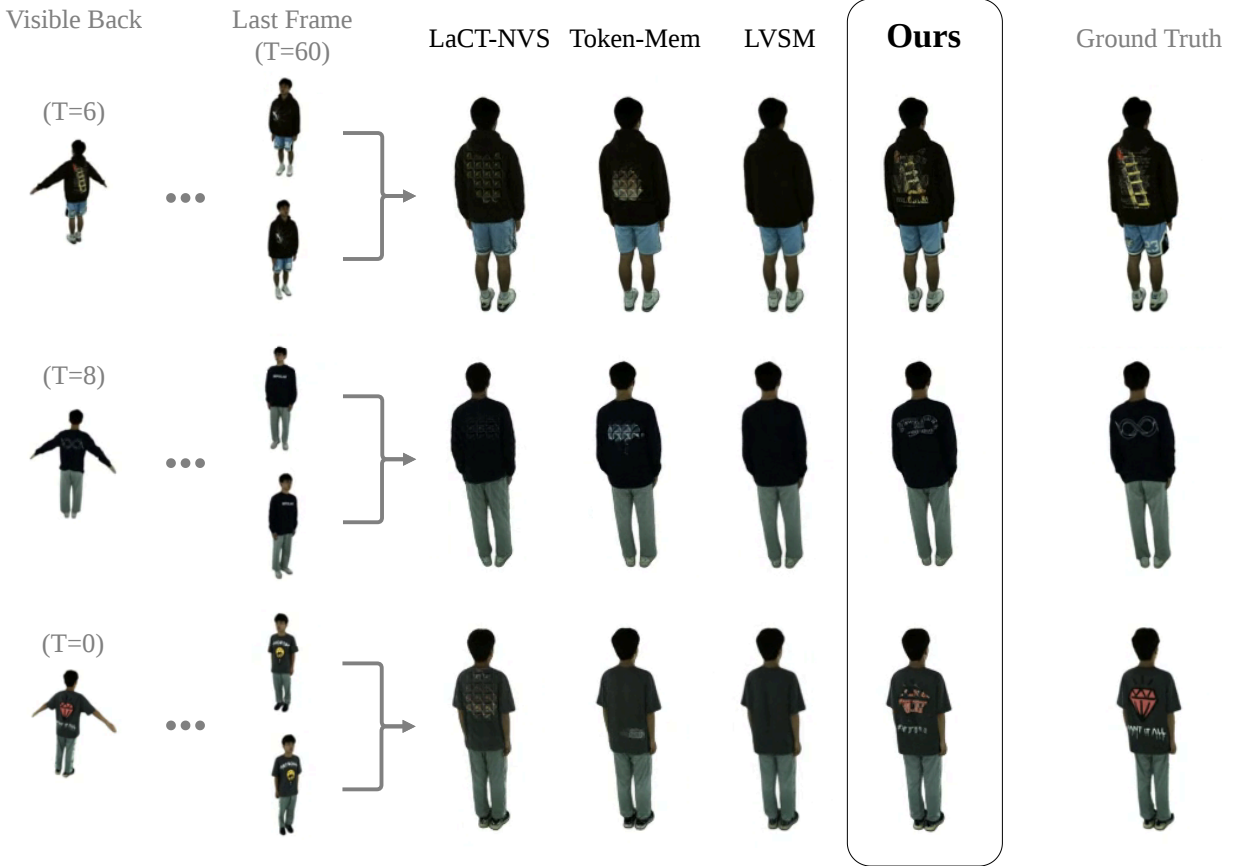


Figure 6. **Memory from Natural Rotations Qualitative Results.** Comparison with LVSM, Token-Mem (CUT3R), and LaCT-NVS on natural video sequences featuring dynamic reveals and occlusions. Left column shows at which timestep the back is revealed. NSTM accurately recalls T-shirt patterns and hair styles. LVSM hallucinates unseen regions, while LaCT-NVS and Token-Mem fail to recall back views. All stateful models are fine-tuned on 24 frames (see Sec. B.1)

Tab. 1 where the model sees the back view only at timestep 0 and needs to remember it while receiving only frontal views over the entire duration of the video. To test, we provide fixed frontal stereo inputs (e.g., `cam15` and `cam00`) and evaluate synthesis as the back naturally reveals and occludes.

Results and Discussions. We present our quantitative comparisons in Tab. 3 and qualitative visual evidence in Fig. 6 measuring performance both averaged across all frames and on the final frame in the ($t = 60$) sequence. The strength

of NSTM remains evident, outperforming Token-Mem and LaCT-NVS by and LaCT-NVS by 1.74 dB and 2.95 dB mPSNR on last frame measurement. Since LVSM is a stateless model the average and last frame performance are identical, demonstrating that it is not able to remember past details that would allow it to increase its accuracy.

B.2. General Novel View Synthesis Test

In addition to the memory stress test in Sec. 4.2 and natural memory test in Sec. B.1, NSTM is fundamentally designed

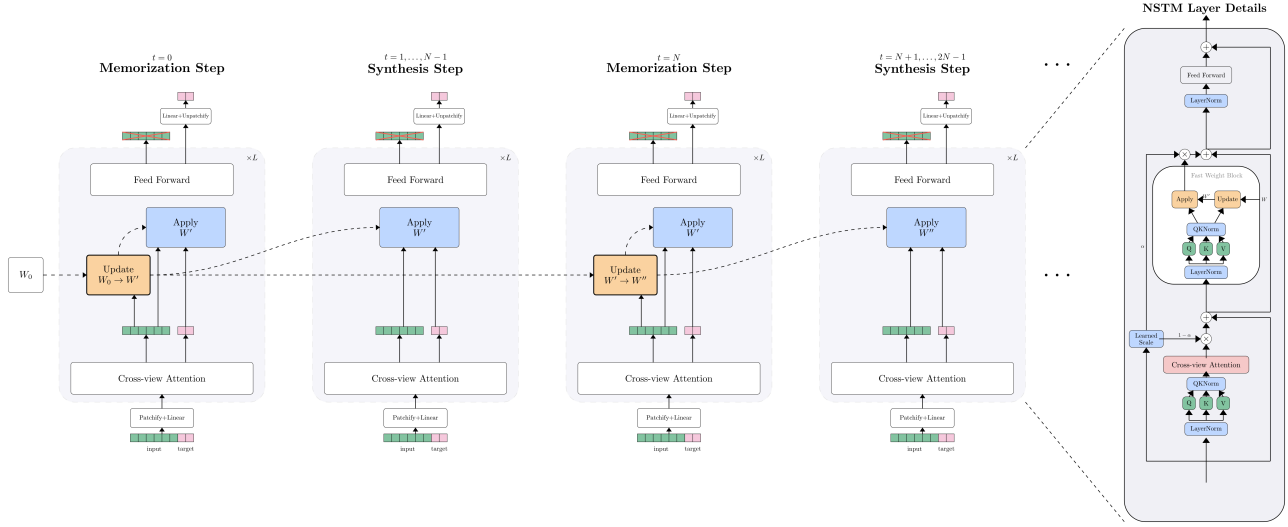


Figure 7. **Online Inference Protocol and NSTM Layer Details.** **Left:** During inference, we run the memorization step periodically while perform the synthesis step per-frame, which enables us to achieve amortized real-time speed. **Right:** We introduce the NSTM Layer consisting of cross-view attention, a fast-weight block and a feed-forward layer. We also incorporate a learned scale that allows the network to balance the contribution of memory and attention to each layer’s output.

Method	Novel View Synthesis Over 60 Timesteps					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	29.48	0.9699	0.0225	0.0917	20.10	0.6509
LaCT-NVS	26.84	0.9567	0.0281	0.0997	17.62	0.5869
Token-Mem	22.81	0.9273	0.0477	0.1180	14.04	0.4595
Ours	29.11	0.9696	0.0223	0.0894	19.93	0.6703

Table 4. **General Novel View Synthesis Comparison.** **best**, **second best**, and **third best** are colored. We use two fixed frontal cameras as the input to NSTM, and use random target cameras at each time step to evaluate the novel view synthesis performance **averaged over 60 timesteps across 360 scenes**.

to be a highly versatile dynamic NVS framework. To evaluate its generalized synthesis capabilities, we conduct experiments under a standard, unconstrained streaming protocol. **Evaluation Protocol.** To simulate typical real-world telepresence or broadcasting scenarios where input camera rigs are fixed, we restrict the streaming inputs to only two static frontal views (e.g., `cam15` and `cam01`). The model is then tasked with rendering the dynamic scene from completely random, time-varying target viewpoints. Because the target viewpoints are completely randomized at every frame, this unconstrained protocol inherently breaks the temporal continuity required for persistent memory accumulation. We utilize this setup strictly as a stress test for per-frame spatial generalization.

Results and Discussions. Taking inputs from only the frontal cameras rarely add helpful temporal context. Despite this protocol lacking meaningful temporal context, Table 4 shows that NSTM consistently outperforms the stateful LaCT-NVS and Token-Mem baselines. Furthermore,

NSTM remains competitive in standard structural metrics (LPIPS, SSIM, and mSSIM) against the purely stateless LVSM baseline, while outperforming on LPIPS and DISTS [11] demonstrating that NSTM’s conditioning on past observations drives higher perceptual-quality synthesis in the occasions that temporal context adds useful context even in temporally randomized NVS scenarios. This demonstrates our model’s spatial rendering capabilities.

C. Online Inference Protocol

Step	Latency (ms) ↓	FPS ↑
Memorization	58.14	17.19
Synthesis	27.01	37.02

Table 5. **Inference Performance and Scaling.** Inference performance breakdown measured on a single H100 GPU with a resolution of 256x256.

As shown in Fig. 7, NSTM explicitly decouples the frequencies of the Memorization Step and the Synthesis Step during online inference **with the measured performance for each step shown in Tab. 5**.

Decoupled Memorization and Synthesis Frequencies. Unlike models that rigidly couple memory updates and queries at every timestep, NSTM operates on independent execution strides. For our evaluation on the MVHumanNet++ benchmark, the sequences provided by the dataset are aggressively subsampled, capturing only 1 frame every

25 frames from the raw 30 FPS videos (i.e., subsampled at 1.2 FPS). Consequently, the temporal gap between consecutive inputs is ~ 833 ms. We train with and provide our evaluation results by memorizing every other frame with these sequences. MVHumanNe++ also provides 30 FPS videos of the sequences and for our supplemental videos we produce our inferred outputs with NSTM by synthesizing 29 frames for every 1 step of memorization.

D. Detailed Model Architecture

Fig. 7 shows our NSTM layer in more details. Different from LaCT-NVS [68], we add an additional learned weight α to balance the contribution from the *cross-view attention* module and the *TTT* memory module.

E. Foreground Matting and Background Augmentation

To facilitate clean isolation of the dynamic subject from the studio background, we output a 4-channel $\text{RGB}\alpha$ image. During training, we employ background augmentation: we composite both the ground-truth foreground and our model’s predicted $\text{RGB}\alpha$ output onto randomly sampled solid color backgrounds before computing the photometric loss. This augmentation forces the network to learn a highly accurate alpha matte. Consequently, at inference time, NSTM produces a clean, disentangled alpha channel, enabling seamless compositing of the synthesized human subject into arbitrary novel environments.